

# Garvit Joshi

## Machine Learning Engineer

📍 Dehradun, Uttarakhand — 📞 +91 7534865942

✉ garvitjoshi46@gmail.com — 🗣 Languages: English, Hindi

 LinkedIn

 /Github

 /Kaggle

 /Leetcode

 Portfolio

## Technical Skills

|                  |                                                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming      | Python, SQL                                                                                                                                          |
| Machine Learning | Scikit-learn, XGBoost, Classification, Regression, Ensemble Learning, Feature Engineering, Model Evaluation, Cross Validation, Hyperparameter Tuning |
| Deep Learning    | PyTorch, TensorFlow, CNNs, Transfer Learning, Neural Networks                                                                                        |
| Computer Vision  | YOLOv8, OpenCV, Object Detection, Image Classification, Data Augmentation                                                                            |
| Data Analysis    | Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Statistics                                                                      |
| Deployment       | FastAPI, Docker, ONNX Runtime                                                                                                                        |
| Tools            | Git, GitHub, Jupyter Notebook                                                                                                                        |

## Professional Experience

### Technical Educator – Robotics — Moonpreneur

Oct 2024 – May 2025

MicroPython • Robotics • STEM Education

- Mentored 60+ students(age 9-13) in Python programming and robotics using BBC micro:bit, emphasizing logical reasoning and structured debugging.
- Conducted 15+ practical programming sessions covering sensors, embedded systems, and robotics fundamentals.
- Guided 20+ students through debugging, algorithm development, and project implementation using MicroPython.
- Managed 6 concurrent learning batches while maintaining consistent instructional quality and technical mentorship.

## Featured Machine Learning Projects

### NeuroGuard – Stroke Risk Prediction

GitHub

Python • Scikit-learn • XGBoost • FastAPI

- Engineered a full-stack ML pipeline for cerebrovascular risk prediction, achieving 87% F1 score on test data.
- Benchmarked 5 models (Logistic Regression, Random Forest, XGBoost, AdaBoost, Naive Bayes) using 5-fold CV.
- Curated 15+ clinical features and applied SMOTE balancing, improving recall by 12% for high-risk cases.
- Deployed a FastAPI inference service with sub-100ms latency for real-time risk assessment.

### Industrial Defect Detection

GitHub

PyTorch • YOLOv8 • OpenCV • ONNX Runtime

- Architected a CV pipeline for automated defect detection, achieving 0.92 mAP on production-grade test sets.
- Applied mosaic augmentation and CLAHE preprocessing, boosting mAP by 8% across varying lighting.
- Optimized YOLOv8 models via ONNX Runtime, achieving 2× faster inference with negligible accuracy loss.
- Evaluated performance using precision, recall, and mAP, iteratively fine-tuning hyperparameters.

### Aegis Council – Multi-Model AI Inference System

GitHub

Python • Transformers • FastAPI

- Created a multi-model inference framework that aggregated responses from LLMs, reducing hallucination errors by 18%.
- Implemented weighted voting and configurable routing strategies to compare outputs across 4+ transformer models.
- Designed a pluggable architecture enabling seamless integration of new models within 2 hours.
- Developed a web-based dashboard for side-by-side model evaluation across 10+ reasoning tasks.

## Education

### Bachelor of Technology in Computer Science

Aug 2021 – Jul 2025

Specialization: Data Science (Artificial Intelligence & Machine Learning)

Lovely Professional University

CGPA: 7.61

## Achievements & Certifications

- Amazon ML Summer School: Top 5% of 63,000+ applicants selected for intensive ML training.
- AWS Deep Racer: Ranked in the top 10% in India for virtual autonomous racing league.
- CodeKaze coding competition: Ranked in the top 17k globally out of 150,000+ participants.